

KUMEJIMA, Japan

WMO: 479290

Lat: 26.3381N

Lon: 126.8039E

Elev: 21

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.0	53.4	36.1	31.2	54.8	38.3	34.0	56.2	21.2	61.6	19.3	61.7	7.7	330	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	90.4	81.3	89.5	81.0	88.6	80.6	82.7	88.1	82.1	87.4	81.5	86.8	9.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.2	162.5	86.3	80.6	158.9	86.0	79.9	155.5	85.5	46.7	88.3	46.0	87.7	45.3	87.0	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.7	18.3	16.2	DB	46.6	91.7	2.2	1.3	45.0	92.6	43.7	93.4	42.4	94.1	40.8	95.0	
			WB	42.7	83.3	2.2	0.9	41.1	84.0	39.8	84.5	38.6	85.1	37.0	85.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	74.0	62.5	63.3	66.1	70.9	75.6	81.1	84.6	84.3	82.2	77.9	72.5	66.1	(6)
(7)		DBStd	8.71	4.73	5.14	5.09	4.25	3.57	3.53	1.87	1.95	2.58	3.13	3.82	4.52	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	292	106	85	51	6	0	0	0	0	0	0	2	42	(9)
(10)		CDD50	8753	388	374	499	626	794	932	1073	1062	965	866	676	498	(10)
(11)		CDD65	3570	29	39	85	182	329	482	608	597	515	401	228	75	(11)
(12)		CDH74	32136	11	26	105	522	1913	4977	7645	7423	5715	3023	713	63	(12)
(13)		CDH80	10452	0	0	2	25	235	1506	3266	3080	1847	457	34	0	(13)
(14)	Wind	WSAvg	8.7	8.9	9.0	9.0	8.7	7.7	8.3	8.4	8.5	8.3	9.2	9.0	9.1	(14)
(15)	Precipitation	PrecAvg	85.8	5.5	5.5	7.2	7.7	10.2	10.9	5.3	7.8	8.9	6.0	5.0	5.0	(15)
(16)		PrecMax	134.3	15.8	17.4	16.9	20.6	23.2	25.4	16.2	24.2	49.7	23.1	14.2	10.0	(16)
(17)		PrecMin	57.5	1.5	1.1	1.8	2.7	1.2	3.0	0.0	1.2	1.3	0.6	1.3	1.5	(17)
(18)		PrecStd	17.3	2.8	3.4	3.5	4.0	5.1	5.1	4.8	5.9	9.3	5.2	2.9	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	75.5	77.1	79.1	82.8	86.5	90.0	91.5	91.9	90.1	87.4	83.1	78.1	(19)	
(20)		MCWB	68.8	70.5	72.7	76.5	79.7	82.1	81.8	81.5	81.1	79.4	76.3	71.5	(20)	
(21)		DB	73.1	74.4	76.6	80.4	84.3	88.6	90.5	90.5	88.7	85.4	80.9	75.8	(21)	
(22)		MCWB	67.1	68.7	70.7	74.9	78.2	81.4	81.4	81.2	80.4	78.0	74.1	69.0	(22)	
(23)		DB	71.2	72.7	74.9	78.7	82.6	87.4	89.6	89.4	87.6	84.0	79.2	74.1	(23)	
(24)		MCWB	65.1	67.4	69.4	73.8	77.0	80.9	81.0	80.8	79.8	76.8	72.9	67.6	(24)	
(25)		DB	69.6	70.9	73.2	77.0	81.0	86.1	88.7	88.4	86.6	82.5	77.8	72.6	(25)	
(26)		MCWB	63.5	65.7	68.0	72.2	75.9	80.4	80.7	80.5	79.0	75.6	71.6	66.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	70.9	71.9	74.0	77.5	80.6	83.3	83.7	83.4	82.7	80.7	77.7	73.6	(27)	
(28)		MCDWB	74.3	75.4	77.5	81.0	84.8	88.1	89.1	88.8	88.1	85.5	81.6	76.9	(28)	
(29)		WB	68.3	70.3	72.3	75.9	79.1	82.3	82.7	82.5	81.6	79.4	75.8	70.8	(29)	
(30)		MCDWB	71.8	73.5	75.7	79.3	83.0	86.9	88.3	88.1	86.9	84.0	79.5	74.2	(30)	
(31)		WB	66.4	68.5	70.7	74.7	78.0	81.5	82.0	81.8	80.8	78.3	74.4	69.1	(31)	
(32)		MCDWB	70.4	71.9	74.1	78.0	81.7	86.1	87.6	87.4	85.9	82.7	78.3	73.2	(32)	
(33)		WB	64.4	66.6	69.0	73.4	77.0	80.8	81.4	81.3	80.0	77.2	72.9	67.1	(33)	
(34)		MCDWB	68.7	70.2	72.5	76.5	80.3	85.2	87.0	86.7	85.0	81.4	77.1	71.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	7.7	8.1	8.7	8.5	8.2	7.4	8.1	8.2	8.2	7.3	7.1	7.4	(35)	
(36)		5% DB	9.8	10.2	10.2	9.4	8.8	7.7	8.9	9.2	8.8	8.4	8.3	9.1	(36)	
(37)		MCWBR	7.5	7.5	7.6	6.3	5.2	3.9	3.9	4.0	4.3	4.5	5.6	6.5	(37)	
(38)		MCDBR	8.6	9.0	9.4	8.7	8.2	7.6	8.3	8.2	8.2	7.6	7.7	8.1	(38)	
(39)		MCWBR	7.8	7.9	8.5	6.8	5.4	4.2	4.1	4.0	4.2	4.4	5.8	7.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.471	0.501	0.575	0.582	0.523	0.482	0.475	0.491	0.488	0.487	0.475	0.468	(40)	
(41)		taud	2.091	2.032	1.868	1.912	2.135	2.311	2.329	2.275	2.252	2.189	2.174	2.104	(41)	
(42)		Ebn at Noon	241	245	234	237	250	259	260	257	254	246	239	236	(42)	
(43)		Edh at Noon	45	51	64	62	50	42	41	43	43	43	41	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	705	889	1126	1345	1523	1695	2082	1899	1640	1267	909	729	(44)	
(45)		RadStd	95	150	122	145	211	156	173	132	142	114	85	80	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.49	N/A	N/A	+0.70	+0.97	+1.08	N/A	N/A	+189	+131	(46)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	+0.79	+0.95	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page